

1. OIML R-76 NAWI Test Report & Compliance Management System (Legal Metrology Laboratories)

Industry Context

Non-Automatic Weighing Instruments (NAWIs) such as retail balances, laboratory scales, platform scales, and weighbridges are widely used in trade, healthcare, agriculture, and industry, where accurate measurement ensures fair transactions and consumer protection. Under the Legal Metrology Act, 2009, every NAWI model must undergo type evaluation in authorized laboratories as per OIML Recommendation R-76 before approval. Today, test reports are prepared manually using paper sheets, isolated spreadsheets, and static Word templates. Engineers compute errors across hundreds of load points by hand, which often leads to:

- Calculation and formula errors
- Transcription mistakes
- Inconsistent report formats
- Delayed model approvals
- Costly re-testing
- Weak audit trails
- Scattered test records

Current reporting processes are manual and error-prone rather than automated and validated. Observations are checked only after testing, so missed load points or out-of-tolerance readings are found late. There is a growing need for an intelligent software system that guides test data entry, calculates errors automatically, evaluates compliance instantly, and generates standardized OIML R-76-2 reports.

Problem Statement

Design and develop a centralized NAWI Test Report & Compliance Management System that automates test data recording, metrological calculations, and report generation for Legal Metrology laboratories.

The system should capture instrument specifications, laboratory environmental conditions, reference weight traceability, and observations from weighing performance, eccentricity, repeatability, tare, and influence factor tests.

Using configurable OIML R-76 rule tables, the system should automatically calculate indication errors, determine permissible errors by accuracy class, evaluate pass/fail status, and generate standardized test reports in PDF and DOCX formats.

The solution should support multiple laboratories while ensuring accuracy, data security, offline data entry, and complete ISO/IEC 17025 audit traceability.

Objectives

The system should:

- Capture instrument and laboratory test data digitally

- Validate entries in real time during testing
- Calculate errors (E, Ec) and permissible limits automatically
- Evaluate pass/fail compliance for every load point
- Generate standardized OIML R-76-2 reports
- Maintain a searchable repository of test histories
- Provide role-based access and immutable audit trails
- Support rule updates when OIML recommendations are revised

Expected Outcomes

The proposed system will:

- Reduce report turnaround time through automation.
- Eliminate calculation and transcription errors.
- Ensure uniform OIML R-76-2 report formatting across laboratories.
- Flag out-of-tolerance readings immediately during testing.
- Improve audit readiness with tamper-evident logs and traceability.
- Speed up model approval for manufacturers and regulators.
- Support scalable deployment across multiple regional laboratories.